1. (15 points) **Backpropagation for autoencoders.** In an autoencoder, we seek to reconstruct the original data after some operation that reduces the data's dimensionality. We may be interested in reducing the data's dimensionality to gain a more compact representation of the data.

For example, consider $\mathbf{x} \in \mathbb{R}^n$. Further, consider $\mathbf{W} \in \mathbb{R}^{m \times n}$ where $m < n$. Then $\mathbf{Wx}$ is of lower dimensionality than $\mathbf{x}$. One way to design $\mathbf{W}$ so that it still contains key features of $\mathbf{x}$ is to minimize the following expression

$$\mathcal{L} = \frac{1}{2} \left\| \underbrace{\mathbf{W}^T \overbrace{\mathbf{Wx}}^{P \in \mathbb{R}^m} - \mathbf{x}}_{K \in \mathbb{R}^n} \right\|^2$$

with respect to $\mathbf{W}$. (To be complete, autoencoders also have a nonlinearity in each layer, i.e., the loss is $\frac{1}{2} \left\| f(\mathbf{W}^T f(\mathbf{Wx})) - x \right\|^2$. However, we'll work with the linear example.)

(a) (3 points) In words, describe why this minimization finds a $\mathbf{W}$ that ought to preserve information about $\mathbf{x}$.

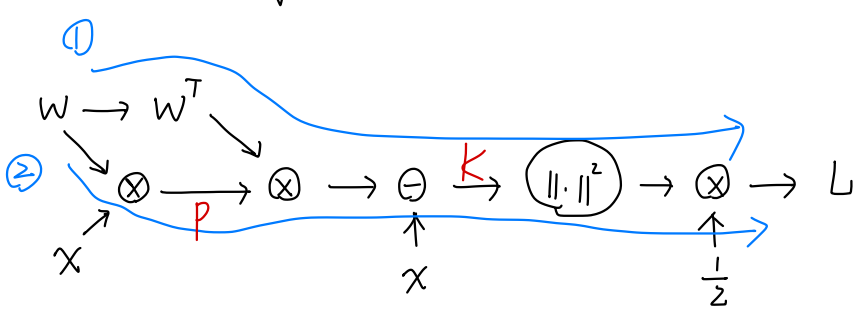
(b) (3 points) Draw the computational graph for $\mathcal{L}$.

(c) (3 points) In the computational graph, there should be two paths to $\mathbf{W}$. How do we account for these two paths when calculating $\nabla_{\mathbf{W}} \mathcal{L}$? Your answer should include a mathematical argument.

(d) (6 points) Calculate the gradient: $\nabla_{\mathbf{W}} \mathcal{L}$.

(a) Minimizing this loss function reduces the error between the original input x and its "reconstruction". This forces the matrix W to capture the most critical features of x in the lower dimentional space so that it can be accurately restored.

(b)



(c) We account for the two parts by "summing" the gradient calculated from each path.

Mathematical Argument: According to the multivariate chain rule, if L depends on W through two intermediate paths/variable (let's call them u, v) the total gradient is the sum of the partial derivatives from each path:

$$\nabla_W L = \frac{\partial L}{\partial u} \cdot \frac{\partial u}{\partial W} + \frac{\partial L}{\partial v} \frac{\partial v}{\partial W}$$

(d)

let $K = W^T W x - x \in \mathbb{R}^n$, $P = Wx \in \mathbb{R}^m$

$$L = \frac{1}{2}\|K\|^2 = \frac{1}{2}K^T K \rightarrow \frac{\partial L}{\partial K} = \frac{1}{2}(2K) = K$$

$$K = W^T P - x \rightarrow \frac{\partial L}{\partial W^T} = \frac{\partial L}{\partial K} \cdot P^T = K(Wx)^T$$

$\Rightarrow$ for the path ① : $\frac{\partial L}{\partial W} = \left(K(Wx)^T\right)^T = (Wx)K^T$

$$\frac{\partial L}{\partial P} = (W^T)^T \frac{\partial L}{\partial K} = WK$$

$\Rightarrow$ for the path ② : $\frac{\partial L}{\partial W} = \frac{\partial L}{\partial P} x^T = WKx^T$

$\therefore$ Total gradient $\nabla_W L = WxK^T + WKx^T$ $\quad (K = W^T W x - x \in \mathbb{R}^n)$

2. (15 points) **Backpropagation for Gaussian-process latent variable model.** An important component of unsupervised learning is visualizing high-dimensional data in low-dimensional spaces. One such nonlinear algorithm to do so is from Lawrence, NIPS 2004, called GP-LVM. GP-LVM optimizes the maximum-likelihood of a probabilistic model. We won't get into the details here, but rather to the bottom line: in this paper, a log-likelihood has to be differentiated with respect to a matrix to derive the optimal parameters.

To do so, we will use apply the chain rule for multivariate derivatives via backpropagation. The log-likelihood is:

$$\mathcal{L} = -c - \frac{D}{2}\log|\mathbf{K}| - \frac{1}{2}\mathrm{tr}(\mathbf{K}^{-1}\mathbf{Y}\mathbf{Y}^T)$$

where $\mathbf{K} = \alpha\mathbf{X}\mathbf{X}^T + \beta^{-1}\mathbf{I}$. To solve this, we'll take the derivatives with respect to the two terms with dependencies on $\mathbf{X}$:

Symmetric

$$\begin{aligned}\mathcal{L}_1 &= -\frac{D}{2}\log|\alpha\mathbf{X}\mathbf{X}^T + \beta^{-1}\mathbf{I}| \\ \mathcal{L}_2 &= -\frac{1}{2}\mathrm{tr}\left((\alpha\mathbf{X}\mathbf{X}^T + \beta^{-1}\mathbf{I})^{-1}\mathbf{Y}\mathbf{Y}^T\right)\end{aligned}$$

Hint: Lawrence states the gradients in his paper, so you can check your answer. To receive full credit, you will be required to show all work. You may find following equation useful:

$$\frac{\partial\mathcal{L}}{\partial\mathbf{K}} = -\mathbf{K}^{-1}\frac{\partial\mathcal{L}}{\partial\mathbf{K}^{-1}}\mathbf{K}^{-1}$$

(a) (3 points) Draw the computational graph for $\mathcal{L}_1$.

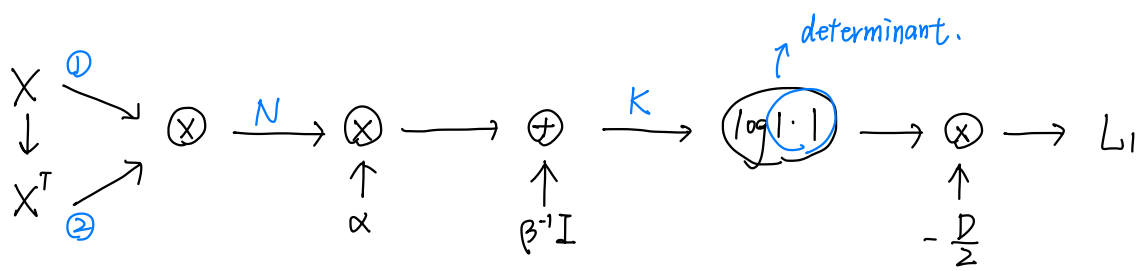
(b) (6 points) Compute $\frac{\partial\mathcal{L}_1}{\partial\mathbf{X}}$.

(c) (3 points) Draw the computational graph for $\mathcal{L}_2$.

(d) (6 points) Compute $\frac{\partial\mathcal{L}_2}{\partial\mathbf{X}}$.

(e) (2 points) Compute $\frac{\partial\mathcal{L}}{\partial\mathbf{X}}$.

(a)



(b)

$$\frac{\partial \ln|\det(X)|}{\partial X} = (X^{-1})^T = (X^T)^{-1}$$

$$Z = XY \rightarrow \begin{cases} \frac{\partial L}{\partial X} = \frac{\partial L}{\partial Z}Y^T \\ \frac{\partial L}{\partial Y} = X^T\frac{\partial L}{\partial Z}\end{cases}$$

$$\frac{\partial L_1}{\partial K} = \frac{-D}{2}\frac{\partial \log|K|}{\partial X} = -\frac{D}{2}(K^T)^{-1}$$

let $N = XX^T \rightarrow \frac{\partial L_1}{\partial N} = \frac{\partial L_1}{\partial K}\cdot\frac{\partial K}{\partial N} = \alpha\frac{\partial L_1}{\partial K} = -\frac{\alpha D}{2}(K^T)^{-1}$

① $\frac{\partial L_1}{\partial X} = \frac{\partial L_1}{\partial N}(X^T)^T = -\frac{\alpha D}{2}\underbrace{(K^T)^{-1}}_{=K}X = -\frac{\alpha D}{2}K^{-1}X$

② $\frac{\partial L_1}{\partial X^T} = X^T\frac{\partial L_1}{\partial N} = -\frac{\alpha D}{2}X^T\underbrace{(K^T)^{-1}}_{=(K^{-1})^T} = -\frac{\alpha D}{2}(K^{-1}X)^T$

$\therefore \frac{\partial L_1}{\partial X} = ① + ②^T$

$= -\frac{\alpha D}{2}(K^{-1}X + K^{-1}X)$

$= -\alpha D K^{-1}X$ #

$$\mathcal{L}_2 = -\frac{1}{2}\mathrm{tr}\left((\alpha\mathbf{X}\mathbf{X}^T + \beta^{-1}\mathbf{I})^{-1}\mathbf{Y}\mathbf{Y}^T\right)$$

(c)

$$\begin{tikzcd}
X \arrow[d] \arrow[dr, "①"] & & & & & & & & & & \\
X^T \arrow[r, "②"'] & \otimes \arrow[r, "N"] & \otimes \arrow[r, "M"] & \oplus \arrow[r, "K"] & \mathrm{inv} \arrow[r, "K^{-1}"] & \otimes \arrow[r, "J"] & \mathrm{tr} \arrow[r] & \otimes \arrow[r] & L_2 \\
 & & \alpha \arrow[u] & \beta^{-1}I \arrow[u] & & YY^T \arrow[u] & & -\frac{1}{2} \arrow[u] &
\end{tikzcd}$$

(d)

$$\frac{\partial\, tr(X)}{\partial X} = I \quad \rightarrow \quad \frac{\partial L_2}{\partial J} = -\frac{1}{2}I \ , \quad I \in \mathbb{R}^{n\times n}$$

$$\frac{\partial L_2}{\partial K^{-1}} = \frac{\partial J}{\partial K^{-1}}\frac{\partial L_2}{\partial J} = YY^T\left(-\frac{1}{2}I\right) = -\frac{1}{2}YY^T$$

use the given hint :

$N = XX^T$

$$\frac{\partial L}{\partial X} = \frac{\partial L}{\partial N}\cdot X$$

$$\frac{\partial L_2}{\partial K} = -K^{-1}\frac{\partial L_2}{\partial K^{-1}}K^{-1} = -K^{-1}\left(-\frac{1}{2}YY^T\right)K^{-1} = \frac{1}{2}K^{-1}YY^TK^{-1}$$

$$\frac{\partial L}{\partial X^T} = X^T\frac{\partial L}{\partial N}$$

① $$\frac{\partial L_2}{\partial X} = \frac{\partial N}{\partial X}\cdot\frac{\partial M}{\partial N}\cdot\frac{\partial K}{\partial M}\cdot\frac{\partial L_2}{\partial K} = \alpha\frac{\partial L_2}{\partial K}X = \frac{\alpha}{2}K^{-1}YY^TK^{-1}X$$

② $$\frac{\partial L_2}{\partial X^T} = \frac{\partial N}{\partial X^T}\cdot\frac{\partial M}{\partial N}\cdot\frac{\partial K}{\partial M}\cdot\frac{\partial L_2}{\partial K} = \alpha X^T\frac{\partial L_2}{\partial K} \overset{②^T}{\Rightarrow} \frac{\alpha}{2}\left(X^TK^{-1}YY^TK^{-1}\right)^T$$

$$= \frac{\alpha}{2}\underbrace{(K^{-1})^T}_{=K^{-1}}YY^T\underbrace{(K^{-1})^T}_{=K^{-1}}X$$

$$\therefore \frac{\partial L_2}{\partial X} = \alpha K^{-1}YY^TK^{-1}X \quad \#$$

(e)

$$\frac{\partial L}{\partial X} = \frac{\partial L_1}{\partial X} + \frac{\partial L_2}{\partial X}$$

$$= -\alpha D K^{-1}X + \alpha K^{-1}YY^TK^{-1}X \quad \#$$

3. (20 points) **Backpropagation through a Residual Block with Biases(Optional for students in C147: Please write 'I am a C147 student' in the solution and you will get full credit for this problem).**

Training very deep neural networks is notoriously difficult due to problems such as vanishing gradients and unstable optimization. A key architectural idea that enables the successful training of modern deep models—including residual networks, Transformers, and large language models (LLMs)—is the *residual (skip) connection.*

In a standard feedforward layer, an input $\mathbf{x}$ is transformed into an output $F(\mathbf{x})$. A residual connection modifies this computation by explicitly adding the input to the output:

$$\mathbf{x} \longrightarrow F(\mathbf{x}) + \mathbf{x}.$$

Rather than learning an entirely new representation at each layer, the model learns a *residual function*—a correction to the input—which significantly improves gradient flow during training.

In this problem, you will analyze backpropagation through a simple residual block that includes both weight matrices and bias terms, and explicitly see how the skip connection alters gradient propagation.

Consider the following network:

Main branch

$$\mathbf{h} = \sigma(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1), \qquad \hat{\mathbf{y}} = \boxed{\mathbf{W}_2\mathbf{h} + \mathbf{b}_2} + \boxed{\mathbf{x},}$$

$h \in \mathbb{R}^d$.

residual Connection

where:

- $\mathbf{x} \in \mathbb{R}^d$ is the input,
- $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{d\times d}$ are trainable weight matrices,
- $\mathbf{b}_1, \mathbf{b}_2 \in \mathbb{R}^d$ are trainable bias vectors,
- $\sigma(\cdot)$ denotes the element-wise sigmoid activation function,
- the term $+\mathbf{x}$ represents a residual (identity) connection.

* key notion :

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial \hat{y}}\left(\frac{\partial \text{Main Branch}}{\partial x} + \underbrace{\frac{\partial x}{\partial x}}_{I}\right)$$

"previous layer" can get at least "I" gradient → prevent vanishing gradient

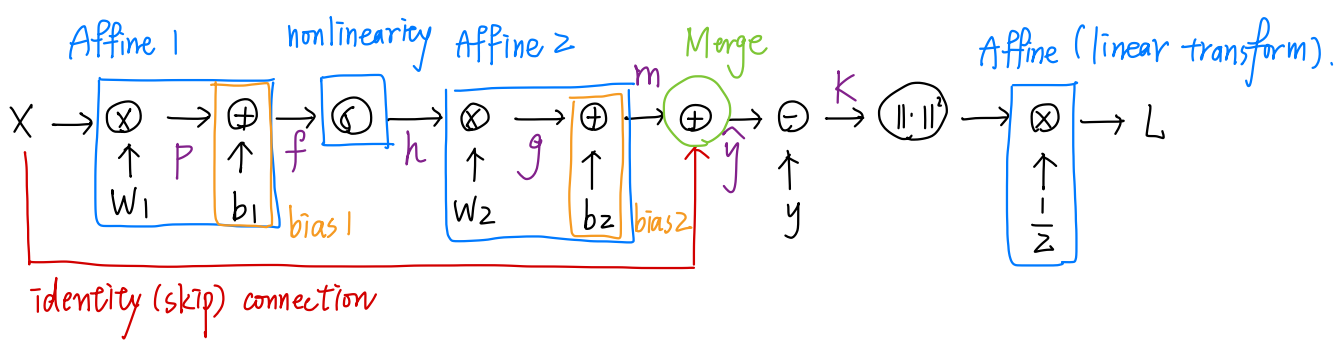
The loss function is the squared error:

$$\mathcal{L} = \frac{1}{2}\|\hat{\mathbf{y}} - \mathbf{y}\|^2.$$

linear (Wx) + Translation (+b)

(a) (4 points) Draw the **computational graph** corresponding to this model and loss. Your graph should clearly indicate:

A • the affine transformation and nonlinearity in the main branch,
B • the identity (skip) connection,
C • the addition operation where the two paths merge,
D • the role of bias terms in each layer.



(b) (5 points) Compute the gradient of the loss with respect to $\mathbf{W}_2$, i.e.,

$$\nabla_{\mathbf{W}_2}\mathcal{L}.$$

Express your answer using matrix/vector notation and clearly show how the error signal propagates from the loss to the output weights.

$$\frac{\partial L}{\partial W_2} = \frac{\partial \hat{y}}{\partial W_2}\frac{\partial L}{\partial \hat{y}}$$

(c) (3 points) Compute the gradient of the loss with respect to $\mathbf{b}_2$, i.e.,

$$\nabla_{\mathbf{b}_2}\mathcal{L}.$$

Briefly explain why the residual connection does or does not affect this gradient.

The residual connection term does not depend on the bias b_2, so its derivative with respect to b_2 is zero.

$\Rightarrow$ It does not affect the gradient

$$L = \frac{1}{2}\|K\|^2 = \frac{1}{2}K^TK \rightarrow \frac{\partial L}{\partial K} = K \in \mathbb{R}^d$$

$$K = \hat{y} - y \rightarrow \frac{\partial L}{\partial \hat{y}} = \frac{\partial K}{\partial \hat{y}}\frac{\partial L}{\partial K} = I \cdot K = K$$

$$\hat{y} = m + x \rightarrow \frac{\partial L}{\partial m} = \frac{\partial \hat{y}}{\partial m}\frac{\partial L}{\partial \hat{y}} = I \cdot K = K$$

$$m = b_2 + g \rightarrow \frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial m} = \frac{\partial L}{\partial g} = K \qquad \therefore \nabla_{b_2} L = K = \hat{y} - y$$

$$g = W_2 h \rightarrow \frac{\partial L}{\partial W_2} = \frac{\partial L}{\partial g} \cdot h^T = K \cdot h^T \in \mathbb{R}^{d \times d} \qquad \therefore \nabla_{W_2} L = K \cdot h^T = (\hat{y} - y)h^T$$

$Z = XY$

$$\frac{\partial L}{\partial X} = \frac{\partial L}{\partial Z} \cdot Y^T$$

$$\frac{\partial L}{\partial Y} = X^T \frac{\partial L}{\partial Z}$$

$$\frac{\partial L}{\partial h} = W_2^T K$$

(d) (5 points) Compute the gradient of the loss with respect to $\mathbf{W}_1$, i.e.,

$$\nabla_{\mathbf{W}_1}\mathcal{L}.$$

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial h}\frac{\partial h}{\partial f} \cdot \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial W_1}$$

In your derivation:

- explicitly apply the chain rule,
- show where the derivative of the sigmoid function appears,
- make clear how the residual connection contributes to the gradient signal reaching $\mathbf{W}_1$.

(e) (3 points) Compute the gradient of the loss with respect to $\mathbf{b}_1$, i.e.,

$$\nabla_{\mathbf{b}_1}\mathcal{L}.$$

derivative of the sigmoid :

$$h = \sigma(f), \quad f = W_1 X + b_1$$

$$\frac{\partial h}{\partial f} = \begin{bmatrix} \frac{\partial h_1}{\partial f_1} & \cdots & \frac{\partial h_d}{\partial f_1} \\ \vdots & \ddots & \\ \frac{\partial h_1}{\partial f_d} & & \frac{\partial h_d}{\partial f_d} \end{bmatrix} = \begin{bmatrix} h_1(1-h_1) & \cdots & 0 \\ \vdots & \ddots & \\ 0 & & h_d(1-h_d) \end{bmatrix}$$

$$= \mathrm{diag}(h \odot (\mathbb{1} - h)) \in \mathbb{R}^{d \times d}$$

$\odot$: element-wise multiplication (Hadamard product)

* For $y = \sigma(x)$, $x, y \in \mathbb{R}$

$$\boxed{\frac{\partial y}{\partial x} = y(1-y)}$$

$$\therefore \frac{\partial L}{\partial f} = \frac{\partial h}{\partial f} \cdot \frac{\partial L}{\partial h} = \mathrm{diag}(h \odot (\mathbb{1}-h)) \cdot W_2^T K$$

$$= (h \odot (\mathbb{1}-h)) \odot W_2^T K$$

$$f = W_1 X + b_1 \ , \ p = W_1 X$$

$$\Rightarrow \frac{\partial L}{\partial f} = \frac{\partial L}{\partial p} = \frac{\partial L}{\partial b_1}$$

$$\therefore \nabla_{b_1} L = h \odot (\mathbb{1}-h) \odot W_2^T K \quad \#$$

$$(K = \hat{y} - y).$$

$$p = W_1 X$$

$$\Rightarrow \frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial p} X^T$$

$$\therefore \nabla_{W_1} L = [h \odot (\mathbb{1}-h) \odot W_2^T K] \ X^T \quad \#$$

Note: The residual connection term x does not contain W_1 so its derivative with respect to W_1 is zero $\Rightarrow$ it vanishes.

P.S. another method:

$$\frac{\partial L}{\partial X} = \frac{\partial P}{\partial X} \cdot \frac{\partial L}{\partial P} = W_1^T \frac{\partial L}{\partial P}$$

use the 3D tensor trick:

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial P} \cdot \frac{\partial P}{\partial W_1} = \frac{\partial L}{\partial P} \cdot X^T$$